



Students ID

Contents

Contents	i
1 Introduction to the model, analysis of the sensor signals and feature extraction	1
1.1 Introduction to the problem and objective	1
1.2 Method	5
1.3 Results	7
2 Anomaly detection and fault classification	9
2.1 Introduction to the problem and objective	9
2.2 Methods	9
2.2.1 Mahalanobis-distance detector	9
2.2.2 Fault classification	10
2.3 Results	11
3 Surrogate modeling	17
3.1 Introduction to the problem and objective	17
3.2 Method	17
3.3 Results	18
4 Model updating by MCMC-MH	21
4.1 Introduction to the problem and objective	21
4.2 Method	21
4.3 Results	24
5 Conclusions	27
Bibliography	29

1 | Introduction to the model, analysis of the sensor signals and feature extraction

1.1. Introduction to the problem and objective

The subject of this digital twinning for health monitoring study case is a triplex single-acting reciprocating machine, as the one shown in Figure 1.1, pumping SAE 50W oil ($\rho = 851.6 \text{ kg/m}^3$, $\nu = 97.63 \text{ mm}^2/\text{s}$ (*cSt*)). This type of pump is used to provide high pressure fluid with 100% duty cycle thanks to its three plungers mounted with 120° offset between each other on the crankshaft. The flow rate fluctuations are reduced when compared to a single or double plunger pump, but it could be reduced even further by using a double-acting triplex pump (higher capital cost).

Some applications of this machine in the industry are: water-jet cutting, oil&gas, CO_2 injection (Carbon Capture and Storage).

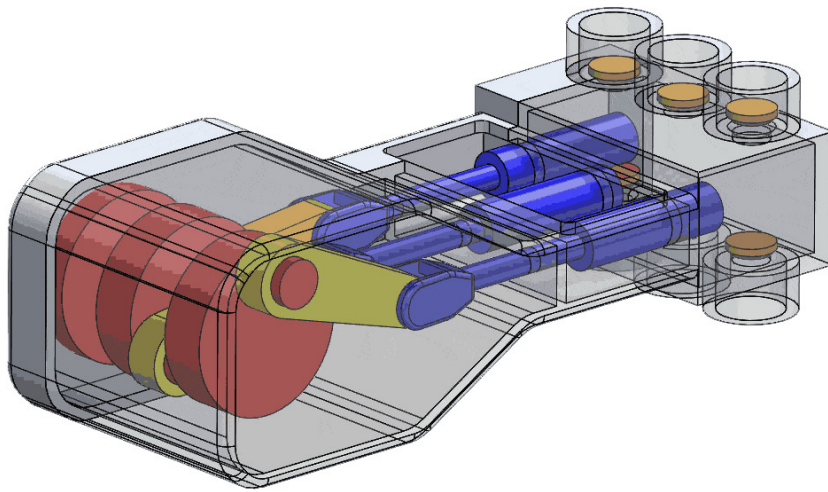


Figure 1.1: CAD model of a triplex reciprocating pump

1| Introduction to the model, analysis of the sensor signals and feature extraction

2

A non-causal model of the triplex pump, the electronics and the hydraulics, based on the Simulink-Simscape model [1] was already modified for this application and provided.

The overview of the model is displayed in Figure 1.2. Starting from the left side of the Simscape model, a 24V DC generator is linked in series with a current sensor, providing the electricity to drive the motor connected to the crankshaft of the triplex pump. The crankshaft target speed is set to around 1000 rpm (corresponding to a given flow rate), a PI controller compares this reference value with the current speed to generate the required torque (at the motor shaft) or equivalently the current. The motor is limited to 5Nm of torque. The theoretical maximum power output required would be: $5[Nm] \cdot (958 \cdot 2) \cdot 2\pi/60 [rad/s] = 1003[W]$ (max motor power 9000W), but it was observed that for the given load and viscous friction forces (medium wear of crankshaft bearings), the motor torque limit acts as bottleneck, not reaching the target speed.

As the volumetric machine has three single-acting plungers, only one inlet and outlet port per cylinder is modelled, controlled by check valves. All the outlet ports are combined with a collector connected with a pressure and flow rate sensor and a 5m long ($\phi = 3 \text{ cm}$) pipe, the same is done for the inlet ports. Finally, a load is modelled as an orifice ($A_{orifice} = 2 \cdot 10^{-4} \text{ m}^2$, $C_{discharge} = 0.7$) and a pressure source of 7 bar to simulate the pressure loss originating from the load. The fluid isn't flowing in a closed loop, in fact the two ends of the pipe are connected to the hydraulic reference block (atmospheric pressure).

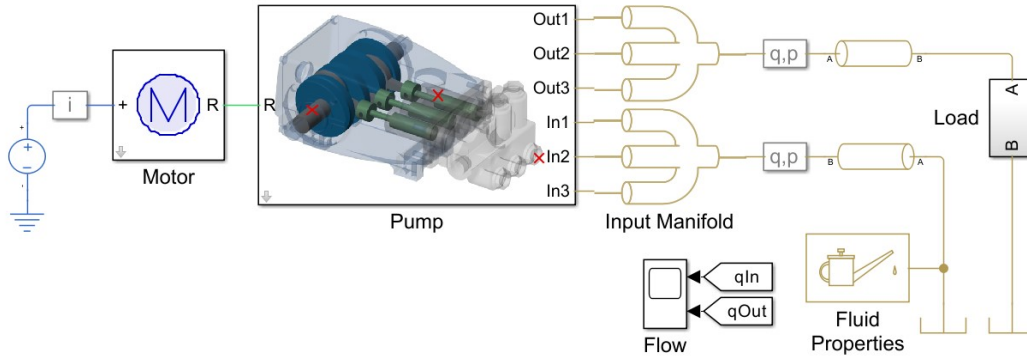


Figure 1.2: Overview of the Simulink-Simscape model of the triplex pump.

The digital shadow of this digital twin is represented by:

- **Current sensor**, measuring the current usage (A) of the motor driving the triplex pump (Figure 1.3a).
- **RPM sensor**, measuring the angular speed (rpm) of the crankshaft, connected

through a gearbox (2:1 gear ratio) with the motor shaft (Figure 1.3b).

- **2 pressure sensors**, measuring the pressure (bar) at the inlet and outlet collector (Figure 1.3c).
- **2 volumetric flow rate sensors**, measuring the flow rate (L/min) of fluid entering and exiting the pump (Figure 1.3c).

To make the measurements realistic, noise was added to the sensor output signals as shown in Figure 1.3.

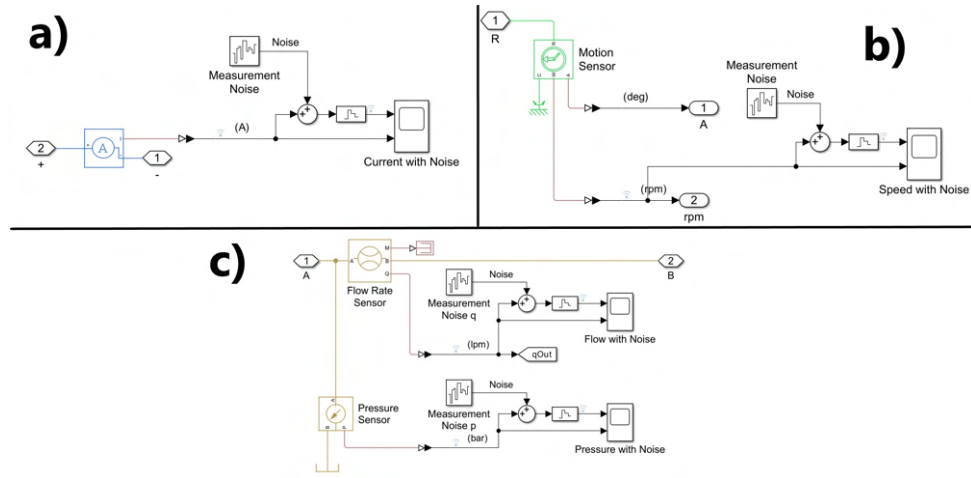


Figure 1.3: Simulink-Simscape models of the sensors.

As the purpose of the digital twin is health monitoring, it should be able to detect anomalies and assess them, three faults were modelled, assuming that a damage assessment was already conducted, selecting these as the most probable type of damage:

- **Bearing fault/wear**, modelled with the variable "bearing_fault_frict_WKSP" which increases the damping coefficient of the revolute joint between the crankshaft and the housing. The undamaged value of damping coefficient is $10^{-4} \frac{N \cdot m}{deg/s}$, while the damping increment variable can go from 0 to $6 \cdot 10^{-4} \frac{N \cdot m}{deg/s}$.
- **Sealing leakage at one plunger head**, modelled with the variable "leak_cyl_area_WKSP" acting on an orifice connected in parallel to the outlet check valve and ending in a hydraulic reference, ranging from $10^{-9} m^2$ to 3.6% of the maximum check valve area (Figure 1.4).
- **Partially blocked inlet to one cylinder**, modelled with the variable "block_in_factor_WKSP" acting on the effective area of the inlet check valve of one cylinder, ranging from 100% to 53% of the inlet check valve maximum area (Figure 1.4).

1| Introduction to the model, analysis of the sensor signals and feature extraction

4

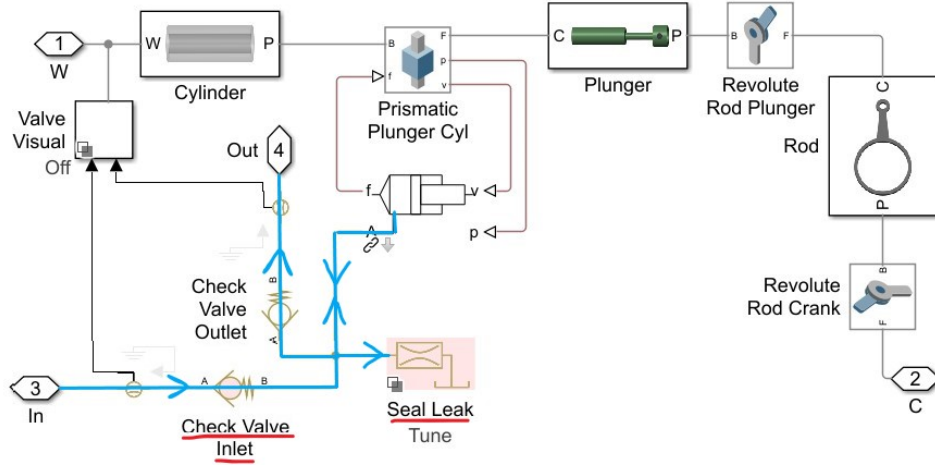


Figure 1.4: Implementation of sealing leakage and fault inlet in the Simscape model of the system (only on 1 plunger). In red are highlighted the faults models, while in light blue the intended path of the fluid.

To understand the sensitivity of the sensors' data to medium-intensity faults, four simulations were completed to study and compare the following scenarios: no faults, medium-sized leak, medium-sized blockage, medium bearing wear.

As later on, the main sensor that will be used is the one measuring the outlet flow rate, the sensitivity of this sensor to the four different scenarios can be appreciated in Figure 1.5.

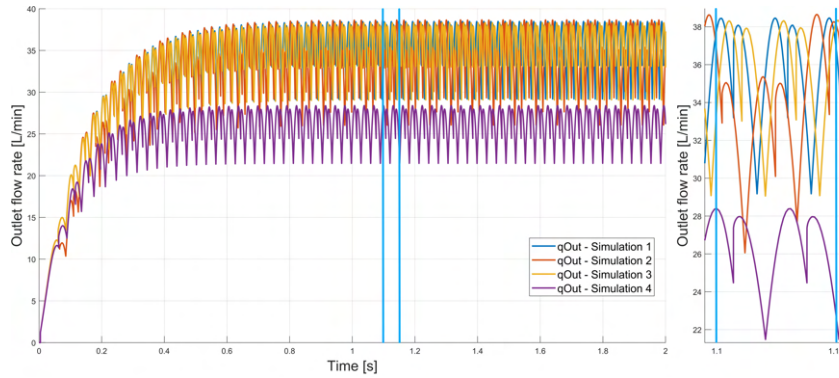


Figure 1.5: Sensitivity of the outlet flow rate sensor to the four chosen scenarios (noise-free signal).

In Figure 1.5, the blue curve represents the outlet flow rate for the **undamaged** triplex pump, it's value oscillates from 29.2 L/min to 38.5 L/min, removing the transient (estimated $t < 1s$), the maximum oscillation around the average (34.1 L/min) is around 14.4% (4.9 L/min). Considering the orange curve, representing the **medium-sized leak** case,

the average value of flow rate after the transient has dropped to 32.85 L/min (-3.7%) as expected, with a larger oscillation of 6.8 L/min. Considering the **medium-sized block-age** scenario (yellow curve), the average value of flow rate remains unchanged compared to the undamaged case, with slightly higher oscillation (5.2 L/min). Finally, the purple curve representing the **medium bearing wear**, shows a drop of 27% (24.9 L/min) in average flow rate compared to the undamaged pump, while maintaining a similar relative oscillation around the mean (14%). The decrease in average flow rate is explained by the larger viscous friction forces at the crankshaft bearings, which leads to an early torsional equilibrium with the 5 Nm applied by the motor, meaning a lower crankshaft speed is reached (648 rpm) compared to the undamaged case (881 rpm).

The capability of the digital twin to describe the damaged behavior of the system, which is one of four main requirements of digital twinning for health monitoring, was already satisfied. The remaining three requirements are:

- stochastic predictions.
- quasi real-time detection and assessment of faults.
- self-updating parameters based on sensor data.

To satisfy the last two requirements is necessary to create a surrogate model that approximates the damage type as well as the fault intensity based on sensors' data. This data unfortunately is time-dependent and includes a transient, so the next step is to extract some features or statistical indicators that can be used also by the surrogate model.

1.2. Method

The steps followed to create the digital twin of the triplex pump for health monitoring are summarized as:

1. Use a proper set of points in the fault space to evaluate the measurement data (Chapter 1).
2. Extract useful features from the time-dependent measurements (Chapter 1).
3. Create a Mahalanobis-distance fault detector based on the features of the undamaged case (Chapter 2).
4. Use the features and the corresponding known types of fault to train an ANN, so that is able to classify all combinations of damage after detection (Chapter 2).

5. Create a surrogate model that can approximate the relation between the fault magnitude and the features of the measurements (Chapter 3).
6. Implement the Bayesian inference through MCMC-MH algorithm to update the surrogate model estimates for faults magnitude based on the measurements' features (Chapter 4).

Many statistical features representing the time-dependent measurements are necessary to distinguish different fault combinations. These features should be characterized by the following metrics: monotonicity, the feature should have a monotonic trend as it approaches failure; prognosability, variability of the feature at failure relative to its total range; trendability, variability of the feature's trend close to failure.

After removing the transient ($t < 0.8s$) from each noisy measurement, computing the signal spectrum, both the time-domain and frequency-domain signals are used to extract the following features:

- frequency of the peak magnitude in the power spectrum.
- power in the low-frequency range (10-20 Hz).
- power in the mid-frequency range (40-60 Hz).
- power in the high-frequency range (100-250 Hz).
- frequency of the spectral kurtosis peak, where the spectral kurtosis is a statistical tool in the frequency domain that can identify non-stationary or non-Gaussian behavior of the measurement.
- range/max variation of the measurement cumulative sum.
- mean value of the measurement.
- variance of the measurement.
- skewness of the measurement, which measures the asymmetry of the data around the mean ($s = \frac{E(x-\mu)^3}{\sigma^3}$).
- kurtosis of the measurement, which measures of how outlier-prone a distribution is ($k = \frac{E(x-\mu)^4}{\sigma^4}$).
- peak-to-peak, the difference between maximum and minimum value of the measurement.
- peak-to-RMS, the ratio of the largest absolute value in the measurement to the root-mean-square (RMS) value of it.

- Root Mean Square (RMS) value of the measurement.
- Mean Absolute Deviation (MAD) of the values in the measurement.

1.3. Results

The four simulations mentioned in Section 1.1 are now used to extract the features from the measurements in both time and frequency domain.

In Figure 1.6, representing the power spectrum of the outlet flow rate measurement, it's possible to notice how the first three scenarios have most of the power focused around the same frequencies (with the exception of the low frequency power peak of the medium-sized leakage case), but with different power amplitudes, while the last one (medium bearing fault) has its power distributed at different frequencies, similarly to what was observed in the time domain.

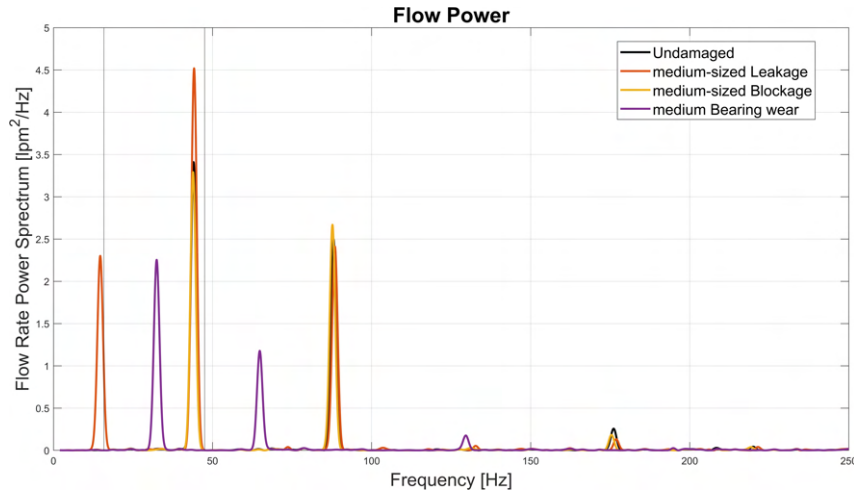


Figure 1.6: Power spectrum of the outlet flow rate for the four different scenarios.

Still focusing on the outlet flow rate measurements, in the following Table 1.1 are summarized the extracted features for the 4 different simulations.

	Undamaged	Case 2	Case 3	Case 4
average [L/min]	35.7	34.3	35.6	26.3
variance [L^2/min^2]	7.6	10.5	7.6	4.8
skewness []	-0.72	-0.44	-0.70	-0.61
kurtosis []	2.79	2.55	2.67	2.91
peak2peak [L/min]	14.3	15.8	13.3	12.2
peak2RMS []	1.15	1.19	1.15	1.19
RMS [L/min]	35.8	34.5	35.7	26.4
MAD [L/min]	2.26	2.63	2.26	1.78
range cum. sum [L/min]	42839	41207	42668	31552
peak power freq. [Hz]	44.03	44.21	43.85	32.46
low freq. power [$L^2/(min^2 Hz)$]	0.58	84.12	0.76	0.57
mid freq. power [$L^2/(min^2 Hz)$]	123.5	161.8	116.5	1.26
high freq. power [$L^2/(min^2 Hz)$]	22.40	22.48	19.67	18.70

Table 1.1: Features extracted from the outlet flow rate measurements for the four chosen scenarios.

2 | Anomaly detection and fault classification

2.1. Introduction to the problem and objective

There are three types of faults (plunger leakage, inlet blockage and bearing wear) that ideally can arise in different components, but for simplicity it's assumed that the faults can only manifest in one component. For example, plunger leakage is modelled only in cylinder number one. Moreover, the only sensor considered, out of 6, is the outlet flow rate.

Before proceeding with the next steps for the creation of a digital twin, it's necessary to generate offline a large set of data that will be used in the process. As it was decided to evaluate 10 fault magnitudes per type of damage (uniformly distributed), the set of evaluating points in the 3D fault space are 1000 in total (curse of dimensionality - $O(c^N)$, with c being the evaluations per dimension, while N the number of problem's dimensions). But to take into account the stochasticity of the features, 50 measurements with different noise are used, increasing the number of evaluation points in the 4-D fault space to $50 \cdot 10^3$. Having created this large dataset ($50000 \times (3 \text{ faults} + 14 \text{ features})$) of features for all combinations of faults, the next goals are: detect faults based on a Mahalanobis-distance threshold and classify faults using a NN.

2.2. Methods

2.2.1. Mahalanobis-distance detector

The Mahalanobis distance M_D is a statistical quantity that allows one to compute how far a point $\bar{f}$ in a N -dimensional space is from a multivariate Gaussian distribution (defined in that same space) given its mean vector $\bar{\mu}_f$ and covariance matrix $[S]$. More precisely defined by the Equation 2.1.

$$M_D = \sqrt{(\bar{f} - \bar{\mu}_f)^T [S]^{-1} (\bar{f} - \bar{\mu}_f)} \quad (2.1)$$

In this application the Mahalanobis distance is used to detect outliers in the multivariate Gaussian distribution of the stochastic features, created offline from the baseline/undamaged condition to recognize faults. The followed procedure to accomplish that is:

1. Given the assumption of Normal multivariate distribution of the features, the Jarque-Bera test is used to identify and remove any distribution of features that doesn't satisfy this kind of PDF, using mainly skewness and kurtosis parameters.
2. Compute the mean vector $\bar{\mu}_f$ and the covariance matrix $[S]$ for the stochastic features in the undamaged case.
3. Choose the threshold value of Mahalanobis distances $M_{D_{th}}$ based on $\bar{\mu}_f$ and $[S]$ of the undamaged case.
4. Compare the $M_{D_{new}}$ of the new measurements, so of new features, with the $M_{D_{th}}$, if $M_{D_{new}} > M_{D_{th}}$ than an anomaly (possibly a fault) is detected.

2.2.2. Fault classification

Once a fault is detected comparing the Mahalanobis distance of the current features with the $M_{D_{th}}$, it's necessary to classify the damage. This classification problem can be solved by approximation with a ANN, trained on the large dataset created offline. The ANN is characterized as follows:

- Epoch size: $49950 \times \text{data_distrib_}\%$ (1000 offline simulations $\times$ 50 random noise additions per simulation, undamaged case excluded).
- Input layer size: 14 features.
- Output layer size: 7 classes (all combinations of faults).
- Number of hidden layers: 1 (not a deep NN).
- Hidden layer size: 10.
- Loss function: cross-entropy function.
- Optimization/training method: scaled conjugate gradient backpropagation (2^{nd} order approximation of loss function).

- ANN architecture: feedforward MLP Neural Network.
- Activation function (hidden layer): tan-sigmoid.
- Activation function (output layer): softmax.
- Data distribution: 70% training, 15% validation, 15% testing.

The procedure followed to train and test the NN was:

1. Import the dataset, split it into training, validation and testing sets.
2. Replace the values of fault with logical values only indicating if there's damage or not, as it's a classification problem not a regression one.
3. Create the output classes using the logical values previously created, the classes are representing the following 7 combinations: leakage, blockage, bearing fault, leakage+blockage, leakage+bearing fault, blockage+bearing fault, all faults.
4. Normalize the inputs, by first computing the average and standard deviation of each column (feature) and then subtract from each column the corresponding mean value and divide it by the corresponding standard deviation.
5. Compute the number of input-output pairs for each class, identify the minimum value and remove randomly pairs from the proper classes so to balance the training dataset.
6. Remove the undamaged cases from both inputs and outputs.
7. Setup the ANN with the specifications just given.
8. Train the ANN until overfitting condition is achieved.
9. Test the ANN performance on the testing set.

As the optimization of the ANN's weights (including biases) is dependent on the initial point in the hyperspace of weights (loss function isn't typically convex, local minima exist), which is randomly initialized, then separate trainings are used.

2.3. Results

Starting from the fault detection, the Jarque-Bera found the distribution of the peak power frequency and the spectral Kurtosis peak to not be a normal pdf, as a consequence these two features were discarded. Figure 2.1 shows the Mahalanobis distance for the

features of the undamaged case, the histogram of its distribution and finally its chosen threshold value: 7.4.

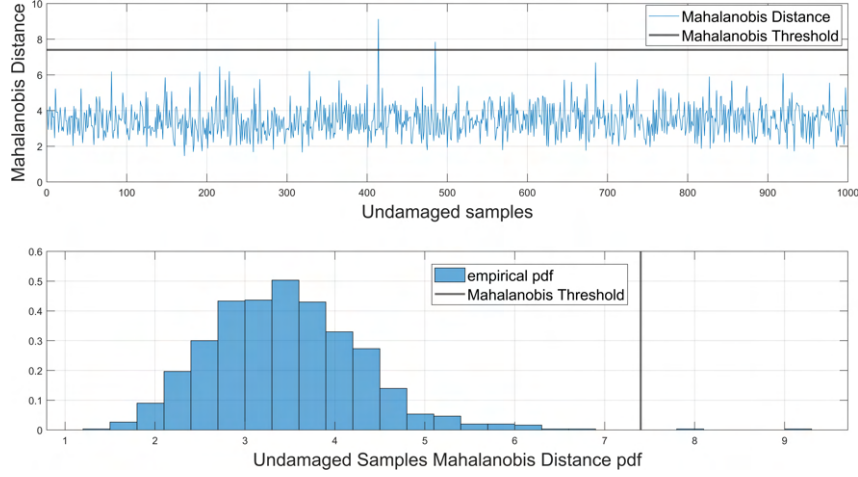


Figure 2.1: Mahalanobis distance for the undamaged features (top). Histogram of Mahalanobis distance distribution (bottom).

The Mahalanobis distance fault detector was tested on small faults hard to detect, precisely: leakage fault parameter of 0.4%, blockage fault parameter of 0.9478 and bearing wear parameter of $6.67 \cdot 10^{-5}$. The results of the test, summarized in Figure 2.2, shows how in the majority of the cases even for small faults the Mahalanobis distance is 3 or more order of magnitude larger than the threshold, detecting all true faults (PoD = 100%), with a false positive rate of 0.1%. Using has the only measurement the outlet flow rate, the fault detector was most sensitive to the bearing fault, followed by the small leakage.

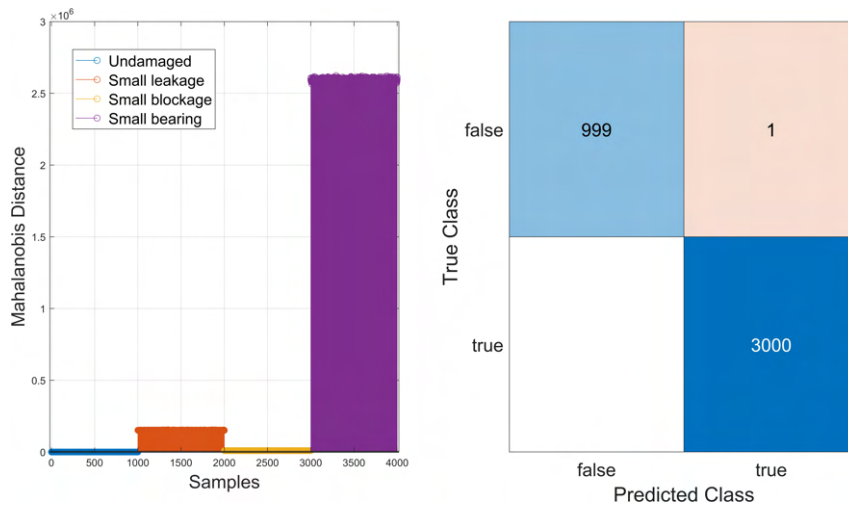


Figure 2.2: Mahalanobis distance of the faulted cases vs the undamaged one (left). Confusion chart (right).

Once the faults can be identified with the Mahalanobis distance detector, the classification of fault is solved by approximation with an ANN, in Figure 2.3 is represented the confusion matrix given by the evaluation of the trained ANN on the testing dataset. This model was trained on an **unbalance dataset**, its classes included all **7 combinations of faults** while the 14 features are **only** including the **outlet flow rate measurements**. As it's noticeable from the figure, the classes describing leakage, bearing, "leakage+bearing" faults are not well classified with the minimum PoD being 4.3%. The arithmetic average of the PoDs was 66.17%.

		Confusion Matrix							
Output Class	Leakage	34 0.5%	0 0.0%	0 0.0%	1 0.0%	0 0.0%	0 0.0%	0 0.0%	97.1% 2.9%
	Blockage	0 0.0%	67 0.9%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Bearing	0 0.0%	0 0.0%	3 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Leakage & Blockage	29 0.4%	1 0.0%	0 0.0%	595 7.9%	0 0.0%	0 0.0%	0 0.0%	95.2% 4.8%
	Leakage & Bearing	0 0.0%	0 0.0%	0 0.0%	0 0.0%	42 0.6%	0 0.0%	2 0.0%	95.5% 4.5%
	Blockage & Bearing	0 0.0%	0 0.0%	66 0.9%	0 0.0%	0 0.0%	556 7.4%	0 0.0%	89.4% 10.6%
	All	0 0.0%	0 0.0%	0 0.0%	0 0.0%	593 7.9%	0 0.0%	5504 73.5%	90.3% 9.7%
	Leakage	54.0% 46.0%	98.5% 1.5%	4.3% 95.7%	99.8% 0.2%	6.6% 93.4%	100% 0.0%	100.0% 0.0%	90.8% 9.2%
		Target Class							
		Leakage	Blockage	Bearing	Leakage & Blockage	Leakage & Bearing	Blockage & Bearing	All	Leakage

Figure 2.3: Confusion matrix, ANN trained on unbalanced dataset, 7 classes and 14 features (outlet flow rate).

To improve the classification capability of the ANN a **balanced dataset** replaced the extremely skewed one used before (around 73.5% of it was including the class "all three faults"). In Figure 2.4 shows how the balanced dataset has improved the classification performance of the ANN, in fact the minimum PoD (for leakage+bearing fault) was 38.9%, with the arithmetic average of PoD being now 72.47%.

		Confusion Matrix						
Output Class	Leakage	51 12.1%	0 0.0%	0 0.0%	15 3.6%	0 0.0%	0 0.0%	77.3% 22.7%
	Blockage	0 0.0%	68 16.2%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Bearing	0 0.0%	0 0.0%	58 13.8%	0 0.0%	0 0.0%	24 5.7%	70.7% 29.3%
	Leakage & Blockage	10 2.4%	0 0.0%	0 0.0%	47 11.2%	0 0.0%	0 0.0%	82.5% 17.5%
	Leakage & Bearing	0 0.0%	0 0.0%	0 0.0%	0 0.0%	21 5.0%	0 0.0%	52.5% 47.5%
	Blockage & Bearing	0 0.0%	0 0.0%	8 1.9%	0 0.0%	0 0.0%	37 8.8%	82.2% 17.8%
	All	0 0.0%	0 0.0%	0 0.0%	0 0.0%	33 7.9%	0 0.0%	46.8% 53.2%
	Leakage	83.6% 16.4%	100% 0.0%	87.9% 12.1%	75.8% 24.2%	38.9% 61.1%	60.7% 39.3%	74.0% 26.0%
		Target Class						
		Leakage	Blockage	Bearing	Leakage & Blockage	Leakage & Bearing	Blockage & Bearing	All

Figure 2.4: Confusion matrix (testing dataset), ANN trained on balanced dataset, 7 classes and 14 features (outlet flow rate).

Next it was added an 8th class representing the undamaged state (not strictly necessary as the Mahalanobis distance is already used for fault detection) and other 14 features (totaling 28) coming from the outlet pressure measurements, as it was observed to have a good sensitivity on all three faults. In this case given the small size (50) of the undamaged cases, the balanced dataset was much smaller (final size of 400×28) than the previous ones, although the number of features was doubled. An advantage of this small scale was the lower computational cost required to simulate and compute the new features for all 400 cases with different fault values (with repetitions but different noise). As displayed in Figure 2.5, the added features improved the classification performance despite the small size of the dataset, reaching an arithmetic average PoD of 77.93% and a minimum value of 57.1%.

It must be noted that the testing dataset is extremely small, a larger one should be created and tested to give statistically meaningful results. To this end, the data regarding fault levels 4 that were discarded during the training of the ANN are now used to compare the current ANN (balanced inputs, 28 features, 8 classes) with the previous one (balanced inputs, 14 features, 7 classes). This comparison is shown in Figure 2.6 and shows similar average PoD (around 50%). Doubling the input features to the ANN, while using a dataset for training 5.6 times smaller has created a fault classifier with similar capabilities.

		Confusion Matrix							
Output Class	Undamaged	5 8.3%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Leakage	0 0.0%	8 13.3%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Blockage	0 0.0%	1 1.7%	8 13.3%	0 0.0%	1 1.7%	0 0.0%	0 0.0%	80.0% 20.0%
	Bearing	0 0.0%	0 0.0%	0 0.0%	5 8.3%	0 0.0%	0 0.0%	3 5.0%	62.5% 37.5%
	Leakage & Blockage	0 0.0%	1 1.7%	0 0.0%	0 0.0%	2 3.3%	1 1.7%	0 0.0%	50.0% 50.0%
	Leakage & Bearing	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	4 6.7%	0 0.0%	40.0% 60.0%
	Blockage & Bearing	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	5 8.3%	100% 0.0%
	All	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	2 3.3%	0 0.0%	80.0% 20.0%
	Undamaged	100% 0.0%	80.0% 20.0%	100% 0.0%	100% 0.0%	66.7% 33.3%	57.1% 42.9%	62.5% 37.5%	57.1% 42.9%
		Undamaged	Leakage	Blockage	Bearing	Leakage & Blockage	Leakage & Bearing	Blockage & Bearing	All
		Target Class							

Figure 2.5: Confusion matrix (testing dataset), ANN trained on balanced dataset, 8 classes and 28 features (outlet flow rate and pressure).

		Confusion Matrix							
Output Class	Leakage	0 0.0%	0 0.0%	0 0.0%	50 14.3%	0 0.0%	0 0.0%	0 0.0%	0.0% 100%
	Blockage	0 0.0%	50 14.3%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Bearing	0 0.0%	0 0.0%	50 14.3%	0 0.0%	0 0.0%	50 14.3%	0 0.0%	50.0% 50.0%
	Leakage & Blockage	50 14.3%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0.0% 100%
	Leakage & Bearing	0 0.0%	0 0.0%	0 0.0%	0 0.0%	50 14.3%	0 0.0%	33 9.4%	60.2% 39.8%
	Blockage & Bearing	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	NaN% NaN%
	All	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	17 4.9%	100% 0.0%
	Leakage	0.0% 100%	100% 0.0%	100% 0.0%	0.0% 100%	100% 0.0%	100% 0.0%	34.0% 66.0%	47.7% 52.3%
	Leakage	Leakage	Blockage	Bearing	Leakage & Blockage	Leakage & Bearing	Blockage & Bearing	All	Leakage
		Target Class							

		Confusion Matrix							
Output Class	Undamaged	14 3.5%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	Leakage	0 0.0%	0 0.0%	0 0.0%	11 2.8%	0 0.0%	0 0.0%	0 0.0%	0.0% 100%
	Blockage	36 9.0%	0 0.0%	49 12.2%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	57.6% 42.4%
	Bearing	0 0.0%	0 0.0%	0 0.0%	48 12.0%	0 0.0%	0 0.0%	34 8.5%	58.5% 41.5%
	Leakage & Blockage	0 0.0%	50 12.5%	1 0.2%	0 0.0%	39 9.8%	0 0.0%	0 0.0%	43.3% 56.7%
	Leakage & Bearing	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	21 5.2%	0 0.0%	42.0% 58.0%
	Blockage & Bearing	0 0.0%	0 0.0%	0 0.0%	2 0.5%	0 0.0%	0 0.0%	16 4.0%	88.9% 11.1%
	All	0 0.0%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	29 7.2%	0 0.0%	42.0% 58.0%
	Undamaged	28.0% 72.0%	0.0% 100%	98.0% 2.0%	96.0% 4.0%	78.0% 22.0%	68.0% 32.0%	68.0% 32.0%	58.0% 42.0%
	Undamaged	Undamaged	Leakage	Blockage	Bearing	Leakage & Blockage	Leakage & Bearing	Blockage & Bearing	All
		Target Class							

Figure 2.6: Confusion matrices of level 4 fault magnitude dataset. Balanced input, 14 features, 7 classes trained ANN (left). Balanced input, 28 features, 8 classes trained ANN (right).

3 | Surrogate modeling

3.1. Introduction to the problem and objective

Now that the digital twin is able to detect faults and classify them, the last step is to predict the fault magnitude based on the experimental measurements (converted into features), which can be used to estimate the RUL (Remaining Useful Life) of the components for example. Achieving this, would satisfy all the main requirements of a digital twin for health monitoring: describing damage processes, giving stochastic predictions, self-updating and quasi real-time predicting.

The estimation of fault magnitude starting from the features of the measurements is implemented via Bayesian inference. In general, to solve this inference numerically (evaluation grid or MCMC-MH) is always necessary to quickly evaluate the predicted measurement (features) for given values of the parameters (faults magnitude in this case). As the relation between the faults magnitude and the measurements is not known explicitly, but a large dataset is available discretely describing it, a surrogate model (ANN) can be trained to swiftly approximate it.

3.2. Method

In this case, the relation between the set of faults magnitude and each feature of the outlet flow rate measurement is approximated separately by a surrogate model, totaling 14. Because the last feature (spectral Kurtosis peak frequency) was not well described by the ANN, it was discarded, resulting in 13 surrogate models.

As the goal is to approximate the relation between faults magnitude and the value of the feature, this is a regression problem solved through ANN. The ANN training consisted in:

1. Passing a set of faults magnitude (level 4 excluded) of the training dataset as input to the ANN.
2. Computing the loss/error function given the current weights of the model and the

expected value of feature.

3. Back-propagating the error according to the level of approximation of the loss function, adding the resulting vector to the previous ones of current epoch.
4. Repeating 1-3 for the entire epoch (if the RAM available is sufficient, if not split into batches).
5. Adding the results of backpropagation obtained from the entire epoch and update the weighs according to the method used.
6. Repeating 1-5 until maximum number of epochs is reached or the validation dataset has shown an increase in its loss function (overfitting).

The ANNs used as surrogate models are characterized as follows:

- Epoch size: $49950 \times \text{data_distrib_}\%$ (1000 offline simulations $\times$ 50 random noise additions per simulation, undamaged case excluded).
- Input layer size: 3 (corresponding to the fault types).
- Output layer size: 1 (corresponding to the feature value).
- Number of hidden layers: 1 (not a deep NN).
- Hidden layer size: 10.
- Loss function: mean squared error function.
- Optimization/training method: Levenberg-Marquardt (2^{nd} order approximation of loss function).
- ANN architecture: feedforward MLP Neural Network.
- Activation function (hidden layer): tan-sigmoid.
- Activation function (output layer): (linear).
- Data distribution: 70% training, 15% validation, 15% testing.

3.3. Results

In Figure 3.1 is shown the regression performance of the surrogate model (ANN) approximating the relation between the faults magnitude and average value of outlet flow rate. More precisely, it's a comparison between the ANN output and the target value of the

feature. The coefficient of determination assumed a value of 0.99997 for this surrogate model and training.

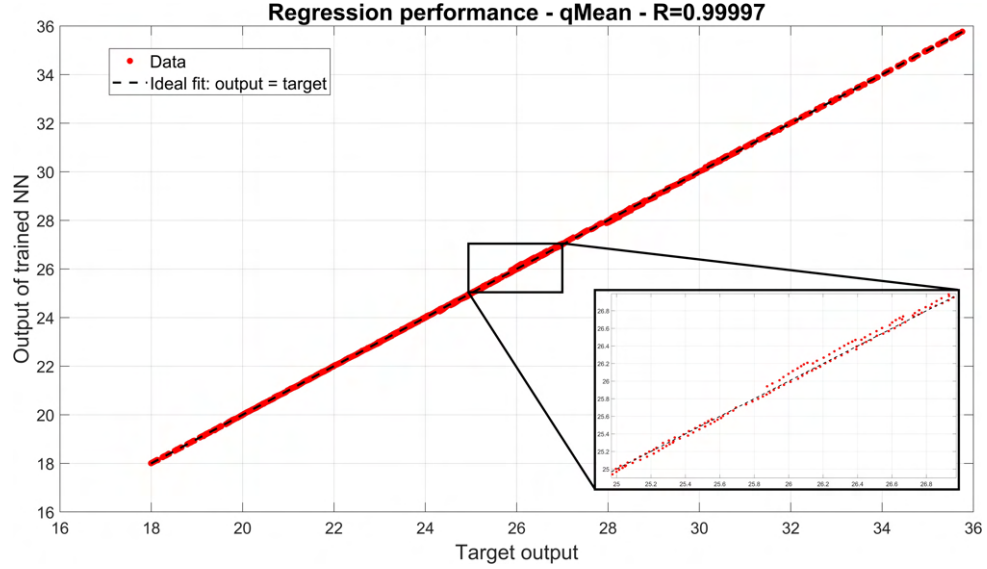


Figure 3.1: Regression performance of ANN (feature: average outlet flow rate), as comparison between NN's output (y-axis) and target output (x-axis).

In the following Figure 3.2 is instead shown the regression performance of the **discarded surrogate model** for feature 14 (spectral Kurtosis peak frequency).

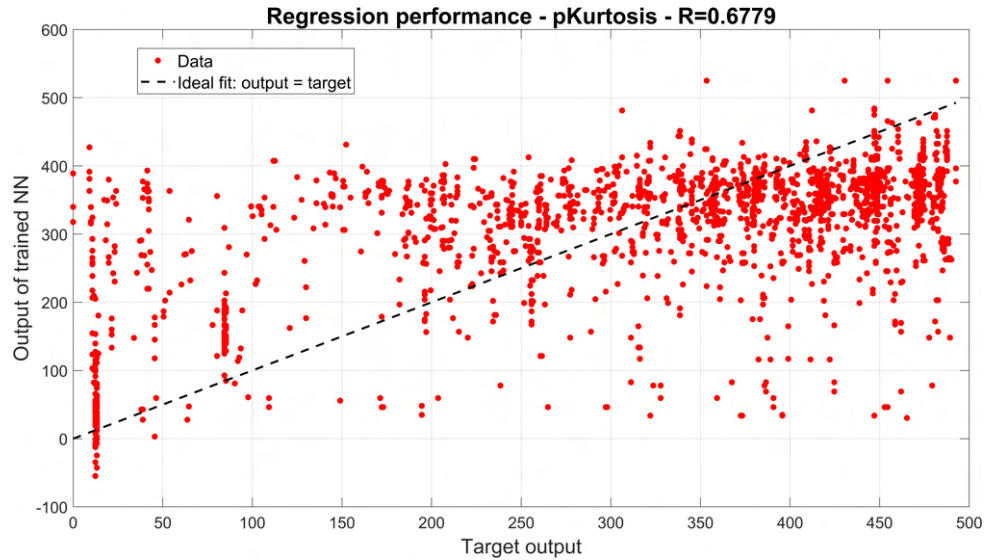


Figure 3.2: Regression performance of ANN (feature: spectral Kurtosis peak frequency), as comparison between NN's output (y-axis) and target output (x-axis).

4 | Model updating by MCMC-MH

4.1. Introduction to the problem and objective

With thirteen surrogate models now enabling the approximation of 'qOut' features based on faults magnitude, the final step involves reversing this relationship to update faults level estimates using observed measurements. This inverse problem is converted into a direct one through Bayesian inference, which is numerically implemented using the Markov Chain Monte Carlo - Metropolis Hastings (MCMC-MH) algorithm.

4.2. Method

The MCMC methods are Monte Carlo methods based on the Markov Chain as the name suggests. The Markov chain is a sequence of states, in this case faults magnitudes, following the Markov property (Equation 4.1).

$$p(\theta_{k+1}|\theta_k, \theta_{k-1}, \dots, \theta_1) = p(\theta_{k+1}|\theta_k) \quad (4.1)$$

Given a sequence of states following this property, there are two possible outcome: random walk and stationary chain. The objective of MCMC method for the implementation of Bayesian inference is to create a stationary Markov chain that can approximate the posterior PDF as well as its definite integral. This method is more efficient than the evaluation grid, as this last one utilizes a uniform computational grid which should be fine enough to describe the narrow likelihood function and wide enough to not miss the posterior PDF.

To create the stationary Markov chain three conditions should be satisfied: irreducibility, positive recurrent and aperiodic. As it's hard to impose such conditions, the reversibility condition (Equation 4.2) can be used instead, which is sufficient for the Markov chain to be stationary.

$$p(\theta_{k+1}|\theta_k) \cdot p(\theta_k) = p(\theta_k|\theta_{k+1}) \cdot p(\theta_{k+1}) \quad (4.2)$$

This condition states: the probability that the process enters in a state θ_{k+1} from θ_k should be the same as it exits θ_{k+1} to enter in θ_k .

The Metropolis-Hastings algorithm is a MCMC method which is able to numerically approximate a posterior PDF by creating a stationary Markov chain describing that distribution. It starts from the reversibility condition, replacing the joint probabilities with the product between a user-defined joint probability and an acceptance ratio α , as well as substituting the prior probabilities of the states with the posterior ones, affected by the observation z (Equation 4.3).

$$q(\theta_{k+1}|\theta_k) \cdot \alpha(\theta_{k+1}|\theta_k) \cdot p(\theta_k|z) = q(\theta_k|\theta_{k+1}) \cdot \alpha(\theta_k|\theta_{k+1}) \cdot p(\theta_{k+1}|z) \quad (4.3)$$

The acceptance ratio $\alpha(\theta_k|\theta_{k+1})$ is 1 as θ_k is the current state. So, the acceptance ratio for θ_{k+1} given the current θ_k becomes:

$$\alpha(\theta_{k+1}|\theta_k) = \frac{q(\theta_k|\theta_{k+1}) \cdot p(\theta_{k+1}|z)}{q(\theta_{k+1}|\theta_k) \cdot p(\theta_k|z)} \quad (4.4)$$

If the user-defined joint distribution is symmetric then $q(\theta_{k+1}|\theta_k) = q(\theta_k|\theta_{k+1})$. Moreover, the posterior probability can be replaced with terms following the Bayes' theorem, obtaining the final form of the acceptance ratio (Equation 4.5).

$$\alpha(\theta_{k+1}|\theta_k) = \min \left(\frac{L(z|\theta_{k+1}) \cdot p(\theta_{k+1})}{L(z|\theta_k) \cdot p(\theta_k)}, 1 \right) \quad (4.5)$$

The upper bound is manually limited to 1 as the probability can't be larger than that.

The MCMC Metropolis-Hastings algorithm for a single parameter can be summarized as follows:

1. Create a prior PDF based on the knowledge of the parameter.
2. Create the user-defined joint distribution q , which will be used to draw new potential states θ_{k+1} from q centered in current state θ_k .
3. Choose an initial state θ_0 of the Markov chain, typically aligned with the prior PDF of θ .

4. Sample θ_{k+1} according to $q(\theta_{k+1}|\theta_k)$.
5. Compute the acceptance ratio $\alpha(\theta_{k+1}|\theta_k)$
6. Draw a sample x from a uniform distribution, if $x < \alpha(\theta_{k+1}|\theta_k)$ then accept θ_{k+1} as new point/state of the Markov chain (equivalent to accepting θ_{k+1} with probability $\alpha(\theta_{k+1}|\theta_k)$).
7. Repeat 4-6 until the total number of points for the Markov chain have been reached.
8. Delete the burn-in (transient) period and apply thinning.
9. Approximate $p(\theta|z)$ and its statistical properties.

In this case only two fault parameters were considered, leakage area and blockage factor, excluding the bearings wear. A problem with the implementation was the extreme difference in range size of two parameters, 10^{-8} (leakage) and 10^0 (blockage). This would make the user-defined function q ineffective for one of the two parameters. The solution was multiplying the first parameter by 10^7 so to bring it to the same order of magnitude of the other one.

The prior distribution chosen for the two faults' magnitude were:

- Scaled leakage area prior Weibull PDF: $f(A_{leakage}) = \frac{1.5}{15} \cdot \left(\frac{A_{leakage}}{15}\right)^{0.5} e^{-(A_{leakage}/15)^{1.5}}$
- Blockage factor "bf" prior beta PDF: $g(bf) = \frac{bf^3(1-bf)^{0.2}}{B(4,1.2)}$, where $B()$ is defined as $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ and Γ is the gamma function $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$, $\Re(z) > 0$.

Assuming the two fault parameters being uncorrelated the joint prior probability density function is shown in Figure 4.1.

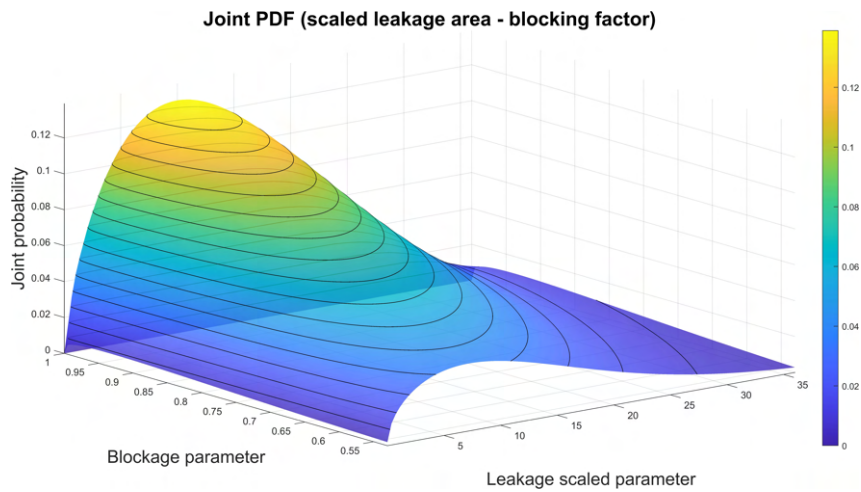


Figure 4.1: Joint PDF of the fault parameters considered.

It's used a Gaussian distribution to describe the likelihood, with variance based on the uncertainty of the feature described: $u = \sqrt{u_{surrogate}^2 + u_{meas}^2}$, where $u_{surrogate}$ is the uncertainty of the surrogate model output based on the faults magnitude and u_{meas} the one of the measurement.

4.3. Results

In Figure 4.2 is shown the comparison between the features' uncertainty coming from the surrogate model performance and the measurement noise. The first one is always more relevant than the second one, as it is generally some order of magnitude larger.

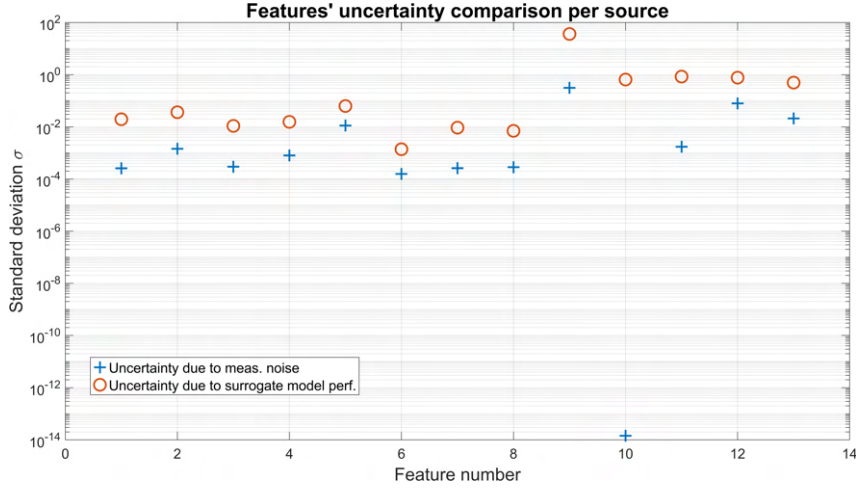


Figure 4.2: Features' uncertainties comparison per type of source.

In Figure 4.3 is displayed the stationary M-H Markov chain approximating the posterior PDF of both analyzed fault parameters. Moreover, in Figure 4.4 are displayed the approximated posterior PDF of the two fault parameters, compared to the true value of fault. The expected value of leakage area approximated by the MCMC-MH algorithm is $1.1865 \cdot 10^{-6}$ while the one of blockage factor is 0.8317. The true values being: $1.2 \cdot 10^{-6}$ and 0.8433. The relative error on the expected value was -1.13% for the leakage area and -1.38% for the blockage factor.

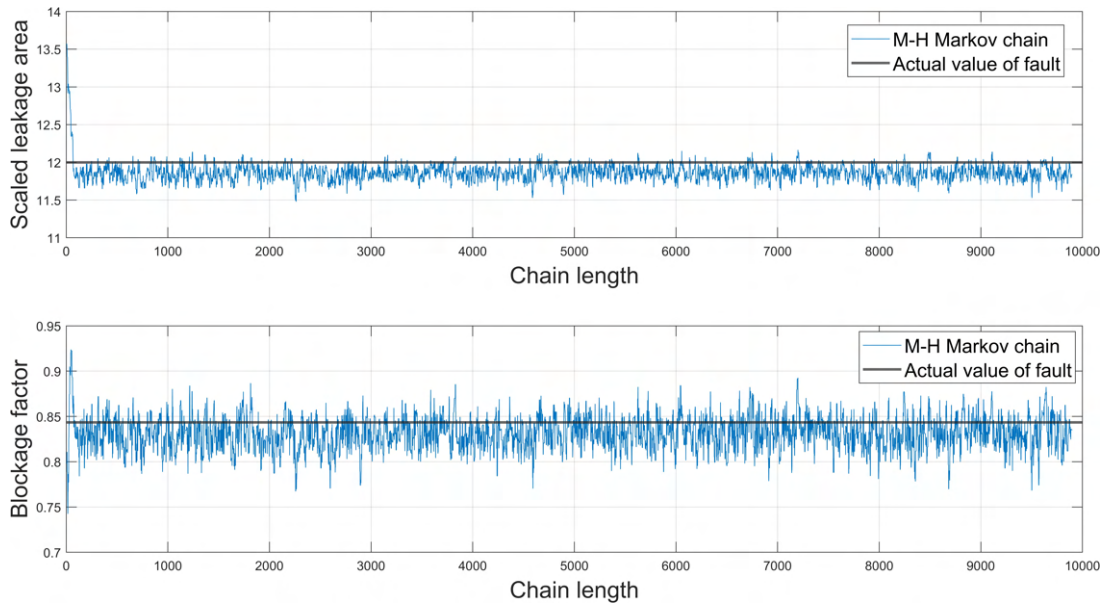


Figure 4.3: Markov chain obtained using M-H algorithm, approximating the posterior PDF of scaled leakage area (top) and blockage factor (bottom).

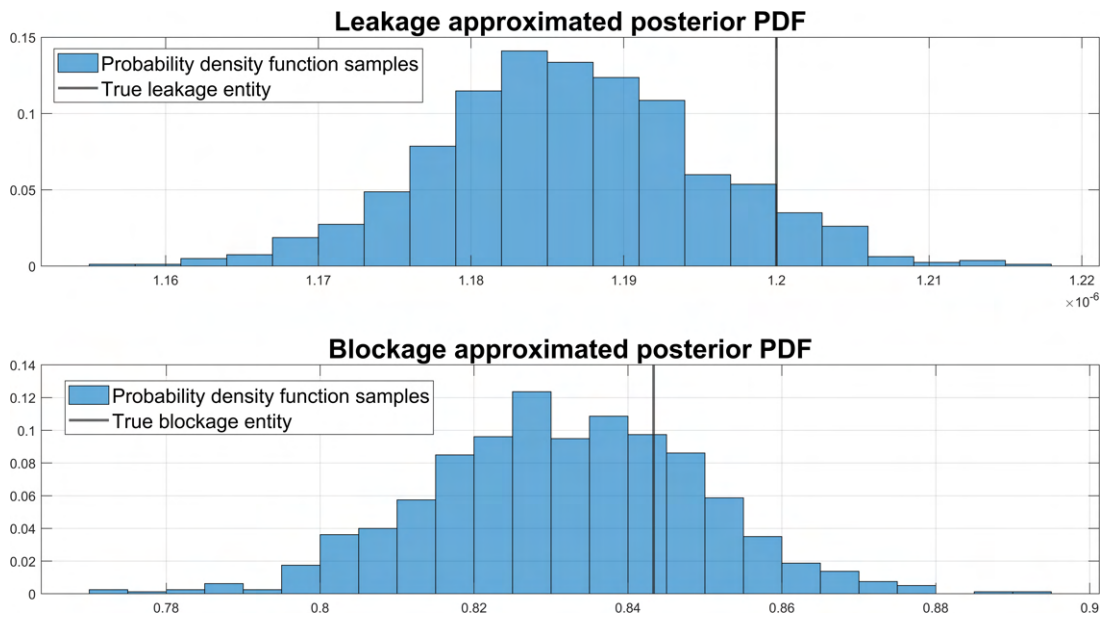


Figure 4.4: Approximated posterior PDF of scaled leakage area (top) and blockage factor (bottom), compared to their true values (gray vertical line).

5 | Conclusions

This application, focused on the creation of a digital twin for health monitoring of a triplex pump, addressed the four critical requirements of digital twinning: describing damage processes, providing stochastic predictions, enabling self-updating parameters, and facilitating quasi real-time fault assessment.

The main achievements were:

- Faults modeling in Simulink-Simscape.
- Mahalanobis distance damage detector, achieving a 100% Probability of Detection (PoD) for even small-scale faults, with a low false positive rate of 0.1%.
- Fault classification via ANN. While initial classification using only outlet flow rate features yielded an average PoD of 66.17%, the inclusion of outlet pressure features and the use of a balanced training dataset improved the average PoD to 77.93%.
- Surrogate modeling the damage magnitude - feature relation using ANNs, obtaining a coefficient of determination of 0.99997 for flow rate averages.
- Faults magnitude updating based on measurements via MCMC-MH algorithm. The algorithm estimated the leakage area and blockage factor with low relative errors of -1.13% and -1.38% respectively, effectively approximating the true state of the system.

The next step would be to include the 5 unused sensors and model damage in all pump's components to get a more complete digital twin of the triplex reciprocating pump, despite the largely increased computational cost.

Bibliography

- [1] MathWorks - Steve Miller. Predictive maintenance in a hydraulic pump. URL <https://www.mathworks.com/matlabcentral/fileexchange/65605-predictive-maintenance-in-a-hydraulic-pump>.

